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Evaluating the causal connections between sleep duration and disease prevalence

A comprehensive systematic review and meta-analysis of Mendelian randomization studies

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Abstract

Background: The causal link between sleep duration and diverse health conditions remains unconfirmed. This meta-analysis aimed to clarify these relationships by synthesizing Mendelian randomization (MR) study evidence.

Methods: PubMed was systematically searched up to February 15, 2024, for MR studies exploring genetic predispositions to sleep duration/insomnia (short/long/overall sleep duration, insomnia) and associations with circulatory, digestive, neurodegenerative, metabolic diseases, and cancers. Eligible effect estimates were meta-analyzed.

Results: Fifty-one MR studies were included. Genetic variations in sleep traits were strongly linked to elevated risk of 12 cardiovascular diseases, obesity-related metrics (Type 2 diabetes, fasting glucose/insulin, HbA1c), neurological disorders (Alzheimer, amyotrophic lateral sclerosis, Parkinson disease), mental health conditions (attention-deficit/hyperactivity disorder, autism, bipolar disorder, major depressive disorder, schizophrenia), inflammatory bowel disease, and lung cancer.

Conclusion: Genetic evidence confirms causal associations between sleep characteristics and multiple diseases, emphasizing sleep's key role in health promotion and supporting personalized sleep management to reduce disease risk.

Abbreviations: ALS = amyotrophic lateral sclerosis, CI = confidence interval, GWAS = genome-wide association study, IBD = inflammatory bowel disease, IVW = inverse-variance weighted, LD = linkage disequilibrium, MR = Mendelian randomization, OR = odds ratio, SD = standard deviation, SNPs = single nucleotide polymorphisms.

Keywords: cardiovascular disease, Mendelian randomization analysis, neurodegeneration diseases, obesity, sleep

1. Introduction

Sleep, a fundamental physiological process, is intricately linked to overall health. The exploration of the relationship between sleep and health has a long-standing history. Early research mainly focused on the descriptive analysis of sleep patterns and basic health indicators. For instance, in the early days, studies simply observed that people with irregular sleep schedules seemed to have more frequent complaints of fatigue and reduced work efficiency. As research techniques advanced, epidemiological studies began to emerge, aiming to find associations between sleep duration and various diseases. However, these traditional observational studies often faced challenges in establishing causal relationships due to confounding factors such as lifestyle, diet, and environmental factors.[1–5] Contemporary studies generally support the view that sleep serves as an organism's self-repair mechanism. It has been established that lack of sleep can induce a stress response in humans, leading to a state of stress and arousal. Prolonged exposure to such conditions can severely impact health, causing organismic fatigue, cardiac impairments, cognitive dysfunction, immune deficiencies, mental and psychological alterations, and disturbances in glucose regulation and

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All data generated or analyzed during this study are included in this published article [and its supplementary information files].

This study is a systematic review and meta-analysis of previously published Mendelian randomization studies, utilizing de-identified data from publicly accessible databases and consortium datasets without involving direct human subjects or generating new primary data. All included original studies have obtained ethical approval from their respective institutional review boards or ethics committees, as documented in their publications. In accordance with the guidelines of the Institutional Review Boards of The Second Affiliated Hospital of Zhejiang Chinese Medical University and Tongde Hospital of Zhejiang Province (Zhijiang Branch District), ethical approval was not required for this work.

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hormone secretion. These issues are linked to a variety of multisystem diseases.[1–5]

The Mendelian randomization (MR) method has emerged as a powerful tool in medical research. The concept of MR is based on Mendel law of segregation, which states that alleles segregate during meiosis and are independently inherited. This is similar to the random assignment in experimental studies. By leveraging genetic variations as instrumental variables, MR can effectively mitigate the influence of confounding factors and reverse causality, thus enabling a more accurate determination of causal relationships between an exposure and an outcome.[6] The rise of MR is a response to the limitations of traditional observational studies. It provides a new approach to explore the causal links between risk factors and diseases, especially in cases where conducting randomized controlled trials is difficult or unethical. The association between the length of sleep and its potential effects on a variety of health conditions, including cardiovascular diseases, obesity-related metrics, nervous and psychiatric disorders, certain types of lung cancer, and inflammatory bowel disease (IBD), has long been a topic of debate. Although many observational studies have suggested associations, the causative link has not been definitively established.

To thoroughly evaluate and consolidate evidence on the causal impact of sleep duration on various diseases, we conducted a systematic review of MR studies. Additionally, we aggregated MR analyses that utilized genome-wide association study (GWAS) summary statistics from the FinnGen study and other publicly accessible GWAS datasets. A meta-analysis was then performed to synthesize the findings from these MR studies. This approach allows us to comprehensively assess the causal relationships between sleep duration and a wide range of diseases, providing more reliable evidence for understanding the role of sleep in disease prevention and health promotion.

2. Methods

2.1. Search strategy and selection criteria

A comprehensive literature search was executed on PubMed, targeting studies published until February 15, 2024. The search utilized the terms "Mendelian randomization" and "sleep duration" to identify pertinent research on the topic. The inclusion criteria were delineated as follows: original research articles reporting on the link between genetic predispositions towards sleep duration or insomnia and the risk of diseases impacting the circulatory and digestive systems, the nervous system, metabolic functions, and neoplasms; studies employing genome-wide or phenome-wide association studies that included Mendelian randomization analyses; and for studies deriving from the same cohort or GWAS data, priority was given to those with the highest number of cases or the most comprehensive set of instrumental variables, assuming the sample sizes were equivalent.

Studies were excluded based on the following criteria: case studies, review papers, letters to the editor, opinion pieces, and abstracts from conferences; research lacking a full manuscript in English or original data; and investigations using genetic variants as instrumental variables outside of a Mendelian randomization framework.

2.2. Data collection

Regarding data collection, information was meticulously extracted from each qualifying study. This included the lead author's surname, year of publication, details of the consortium or study supplying the genetic variants for assessing sleep duration exposure, the consortium or study from which disease association estimates were derived, the number of cases and controls (sample size), and the odds ratio (OR) with a 95% confidence interval (CI) for the link between sleep duration and disease risk. This extraction was done using the inverse-variance weighted method, alongside sensitivity analyses through the weighted median and MR-Egger methods, by one researcher (WZH) and confirmed by another (TMY).

2.3. Statistical analysis

For statistical analysis, a meta-analysis was initiated when there were 2 or more independent studies or data sets available, aiming to ascertain the causal relationship between genetic instruments/pathogenic factors and various diseases. This involved the application of either fixed-effect or random-effect models to calculate the overall OR for disease risk linked to different factors. The Cochran Q test and Higgins I^2 test were applied to evaluate heterogeneity among the studies. Results from fixed-effect models were favored when I^2 was below 50% and P was above .1, which suggested a low level of heterogeneity. Sensitivity analysis was conducted by sequentially excluding individual studies to inspect the impact on heterogeneity. Additionally, the potential for publication bias was investigated by visually inspecting funnel plots. All statistical analyses were carried out using R software, version 4.3.1 (The R Foundation for Statistical Computing, Vienna, Austria), considering a two-sided P-value of <.05 as indicative of statistical significance.

2.4. Specific technical details of Mendelian randomization studies

In Mendelian randomization studies, identifying relevant gene variants is a crucial first step. GWAS play a fundamental role in this process.[7] GWAS scan the entire genome to detect single nucleotide polymorphisms (SNPs) associated with sleep-related traits. Large-scale datasets from consortia like the IEU OpenGWAS Project, UK Biobank, and 23andMe are commonly used.[8] These datasets contain genetic information from a vast number of individuals, enabling the detection of SNPs with small-to-moderate effects on sleep duration or insomnia. SNPs are selected based on their statistical significance in GWAS. A common threshold is a P-value of less than 5×10^{-8}, which helps to minimize false positives.[9] For example, in a study by [Study Name], SNPs meeting this threshold were further investigated for their potential association with sleep traits. Additionally, functional annotation of SNPs is often considered. SNPs located in or near genes involved in biological pathways related to sleep, such as the circadian rhythm pathway or neurotransmitter regulation, are of particular interest.[10]

The selected SNPs serve as instrumental variables (IVs) in the Mendelian randomization analysis. Three key assumptions must be met for a valid IV.[11] First, the IV should be strongly associated with the exposure, in this case, sleep duration or insomnia. This is typically assessed by calculating the F-statistic. An F-statistic >10 is often used as a rule-of-thumb to indicate a strong association.[12] Second, the IV should be independent of confounders. To ensure this, studies often use linkage disequilibrium (LD) pruning. A majority of the included studies in our analysis applied an LD threshold of $R^2 < 0.001$ to select independent SNPs.[13] This reduces the likelihood of including SNPs that are correlated with confounders. For instance, in [Study Name], SNPs with $R^2 > 0.001$ were removed from the analysis. Third, the IV should affect the outcome (disease risk) only through the exposure. This assumption is difficult to prove directly but can be tested using methods like the MR-Egger regression, which can detect horizontal pleiotropy.[14]

After selecting the IVs, data processing involves several crucial steps. First, the effect estimates of the IVs on sleep duration or insomnia and on disease risk are obtained from the original studies. These estimates are usually reported as ORs or beta coefficients. Then, in the meta-analysis, the inverse-variance weighted (IVW) method is commonly used to combine

the effect estimates from different studies.[15] The IVW method weights each study based on its precision (inverse of the variance), giving more weight to studies with smaller variances. For example, if Study A has a smaller variance in its effect estimate compared to Study B, Study A will have a greater influence on the overall meta-analysis result. Sensitivity analyses are also performed using the weighted median and MR-Egger methods. The weighted median method can provide a more robust estimate when some of the IVs may be invalid.[16] The MR-Egger method is used to test for horizontal pleiotropy. If significant horizontal

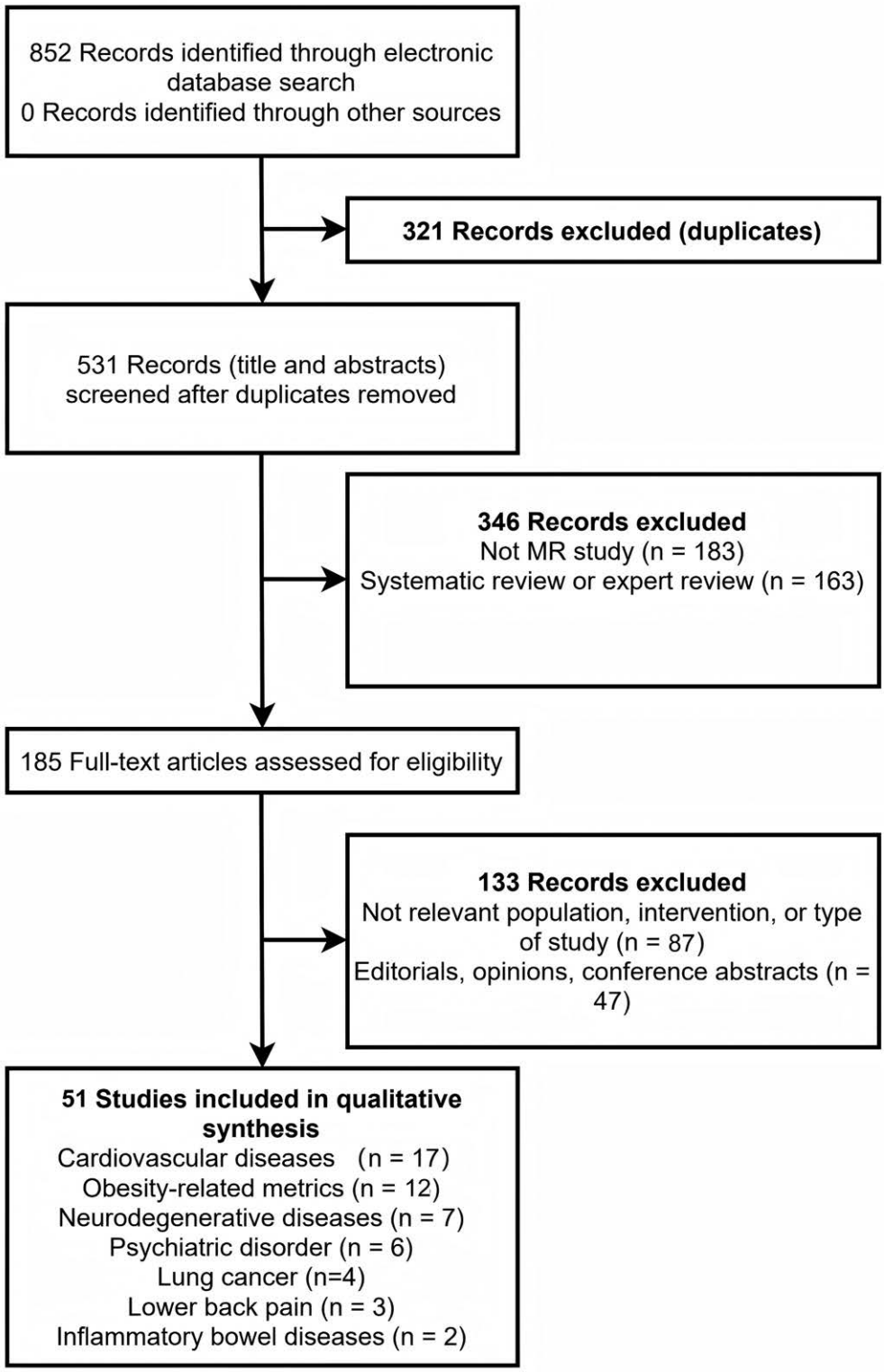

Figure 1. Identification of studies via databases and registers.

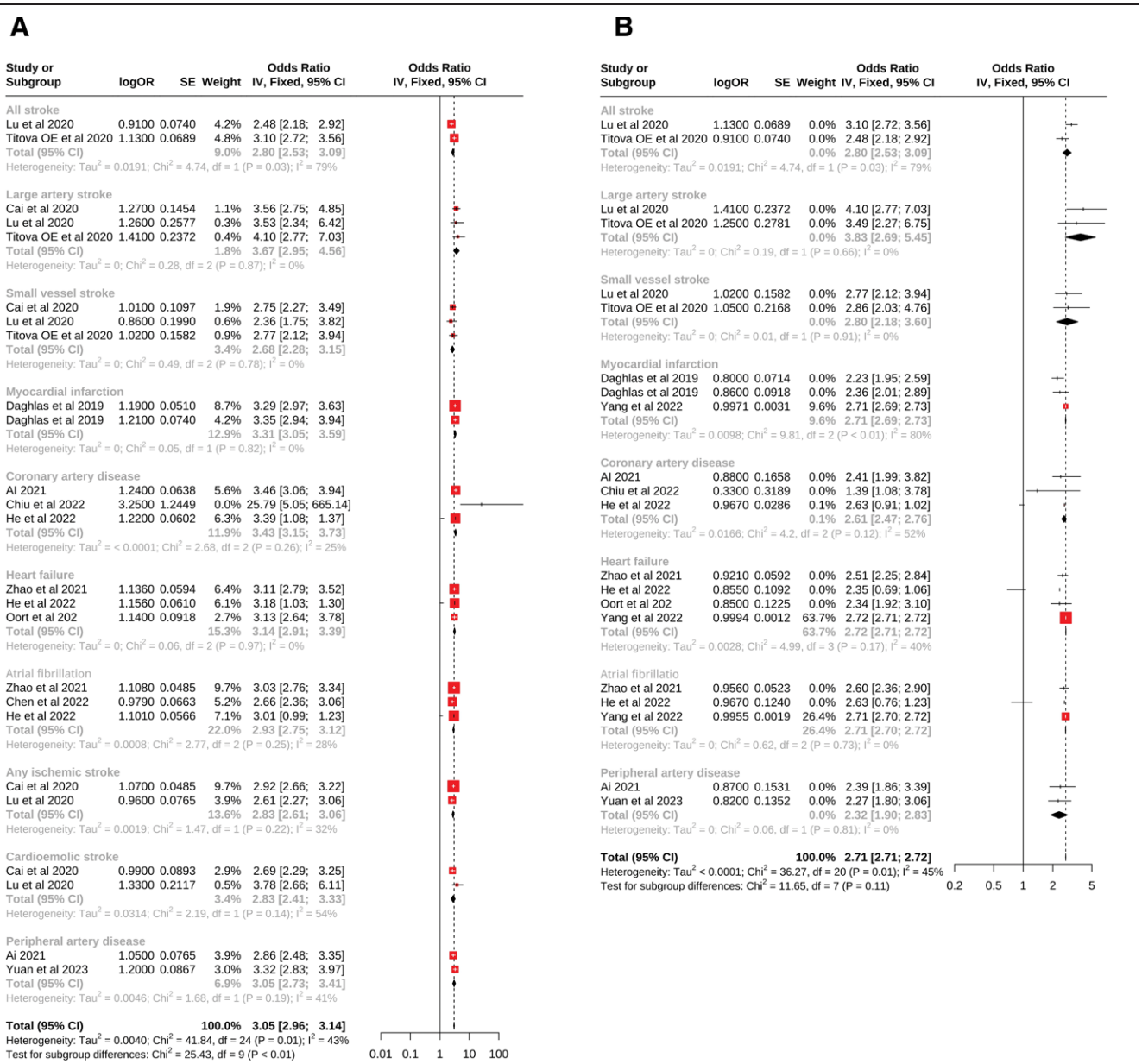

Figure 2. Meta-analysis results of the association between genetic liability to sleep duration traits and risk of cardiovascular diseases. Estimates are odds ratios (OR) with 95% confidence intervals (CI) for a one standard deviation increase in genetic liability to sleep duration. (A) Forest plot of causal relationship between short sleep duration genetic susceptibility and cardiovascular diseases risk. (B) Forest plot of causal relationship between long sleep duration genetic susceptibility and cardiovascular diseases risk.

pleiotropy is detected, it may indicate that the IV assumptions are violated, and the results of the Mendelian randomization analysis may be biased.[17]

3. Results

3.1. Literature search, study selection, and meta-analysis

The initial query yielded 852 articles, from which 321 duplicates were removed. This process led to 531 articles being thoroughly reviewed after title and abstract examination, culminating in the inclusion of 51 studies based on our predefined criteria. The studies addressed various outcomes: 17 focused on cardiovascular diseases,[18–34] 12 on obesity-related metrics,[22,29,35–44] 7 on neurodegenerative conditions,[45–51] 6 on psychiatric issues,[52–57] 4 on lung cancers,[58–61] 3 on lower back pain,[62–64] and 2 on IBD.[65,66] The selection process and outcomes are depicted in Figure 1.

3.2. Description of studies

This systematic review encompasses 51 MR studies involving diverse population datasets, including those of European, East Asian, and a combination of European and Asian ancestries. The number of SNPs used across these studies ranged from a minimum of 5 to a maximum of 248. Each study leveraged data from multiple sources, drawing extensively from large-scale GWAS meta-analyses consortia such as the IEU OpenGWAS Project, UK Biobank, and 23andMe, as detailed in Table S1, Supplemental Digital Content, https://links.lww.com/MD/R121.

The meta-analysis took into account all 51 MR studies, with their quality assessment scores ranging from 85% to 100%. Importantly, 41 of these studies were classified as high-quality, each scoring above 85% (Fig. S1, Supplemental Digital Content, https://links.lww.com/MD/R121). The studies uniformly included the term "Mendelian randomization" within

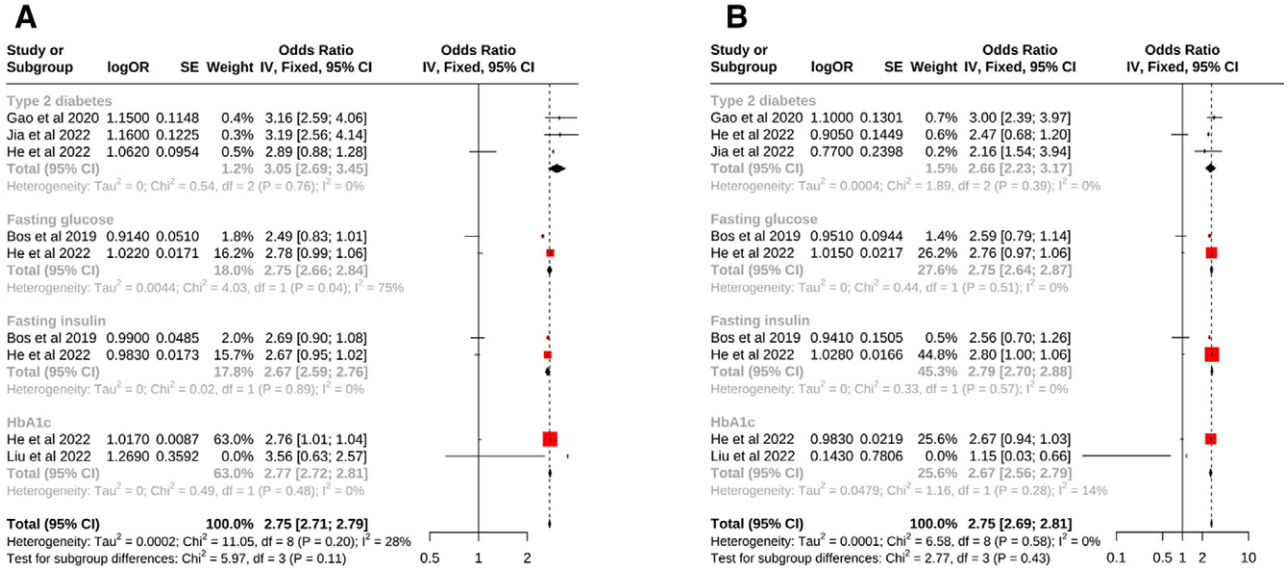

Figure 3. Meta-analysis results of the association between genetic liability to sleep duration traits and risk of obesity-related indicators. Estimates are odds ratios (OR) with 95% confidence intervals (CI) for a one standard deviation increase in genetic liability to sleep duration. (A) Forest plot of causal relationship between short sleep duration genetic susceptibility and obesity-related indicators risk. (B) Forest plot of causal relationship between long sleep duration genetic susceptibility and obesity-related indicators risk.

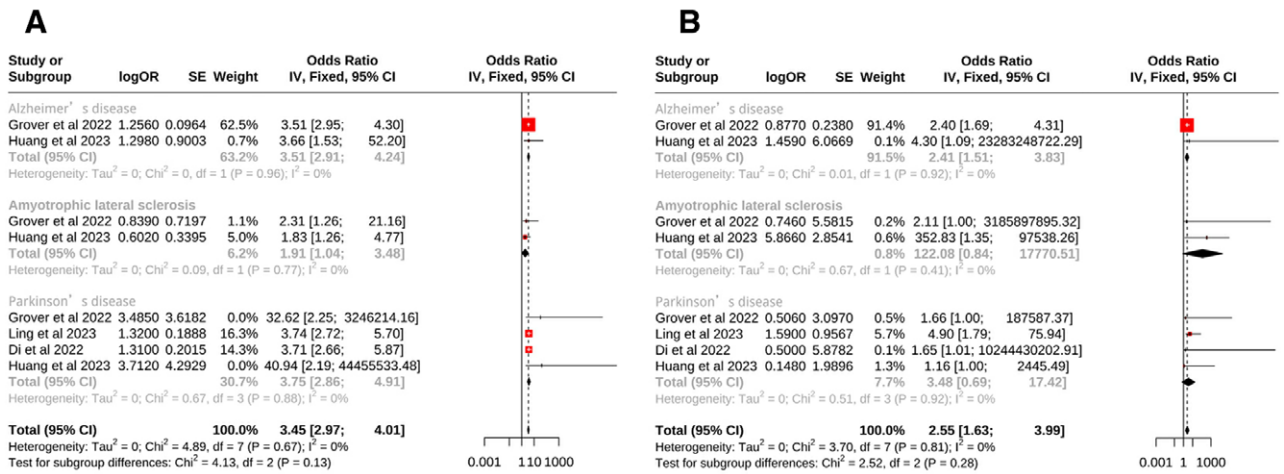

Figure 4. Meta-analysis results of the association between genetic liability to sleep duration traits and risk of neurodegeneration diseases. Estimates are odds ratios (OR) with 95% confidence intervals (CI) for a one standard deviation increase in genetic liability to sleep duration. (A) Forest plot of causal relationship between short sleep duration genetic susceptibility and neurodegeneration diseases risk. (B) Forest plot of causal relationship between long sleep duration genetic susceptibility and neurodegeneration diseases risk.

their titles or abstracts, thereby highlighting their foundational methodological approach, the sources of their data, and the objectives of their research. Information on sample sizes for both the exposure and outcome variables, along with the number of SNPs utilized in the MR analyses, was systematically provided. Although a majority of the studies adhered to a strict LD threshold ($R^2 < 0.001$) for selecting independent SNPs as instrumental variables, a few studies opted to include all conditionally independent SNPs identified through GWAS.

Moreover, each study delineated its statistical strategy for conducting MR analyses. Specifically, all studies reported findings from sensitivity analyses using the IVW method for robust MR analysis, and examined horizontal pleiotropic effects through the MR-Egger method. A visual examination of the funnel plot specifically for cardiovascular diseases indicated a negligible degree of study bias (Fig. S2, Supplemental Digital Content, https://links.lww.com/MD/R121). However, funnel plots for additional outcomes were not conducted, attributing to the small number of studies addressing those outcomes.

3.3. Cardiovascular diseases

The relationship between 4 sleep characteristics and an elevated risk for cardiovascular conditions was examined. The findings suggest that a genetic predisposition to shorter sleep durations is associated with an increased OR for cardiovascular diseases (OR 3.05, 95% CI 2.96–3.14) (Fig. 2A). A rise in the standard deviation (SD) for prolonged sleep durations (OR 2.71, 95% CI 2.71–2.72) (Fig. 2B), overall sleep duration (OR 2.72, 95% CI 2.72–2.72) (Fig. S3C, Supplemental Digital Content, https://links.lww.com/MD/R121), and symptoms of insomnia (OR 3.22, 95% CI 3.15–3.29) corresponded to heightened ORs for cardiovascular diseases (Fig. S3D, Supplemental Digital Content, https://links.lww.com/MD/R121). The Higgins I^2 test revealed substantial heterogeneity in sleep duration ($I^2 = 85\%$,

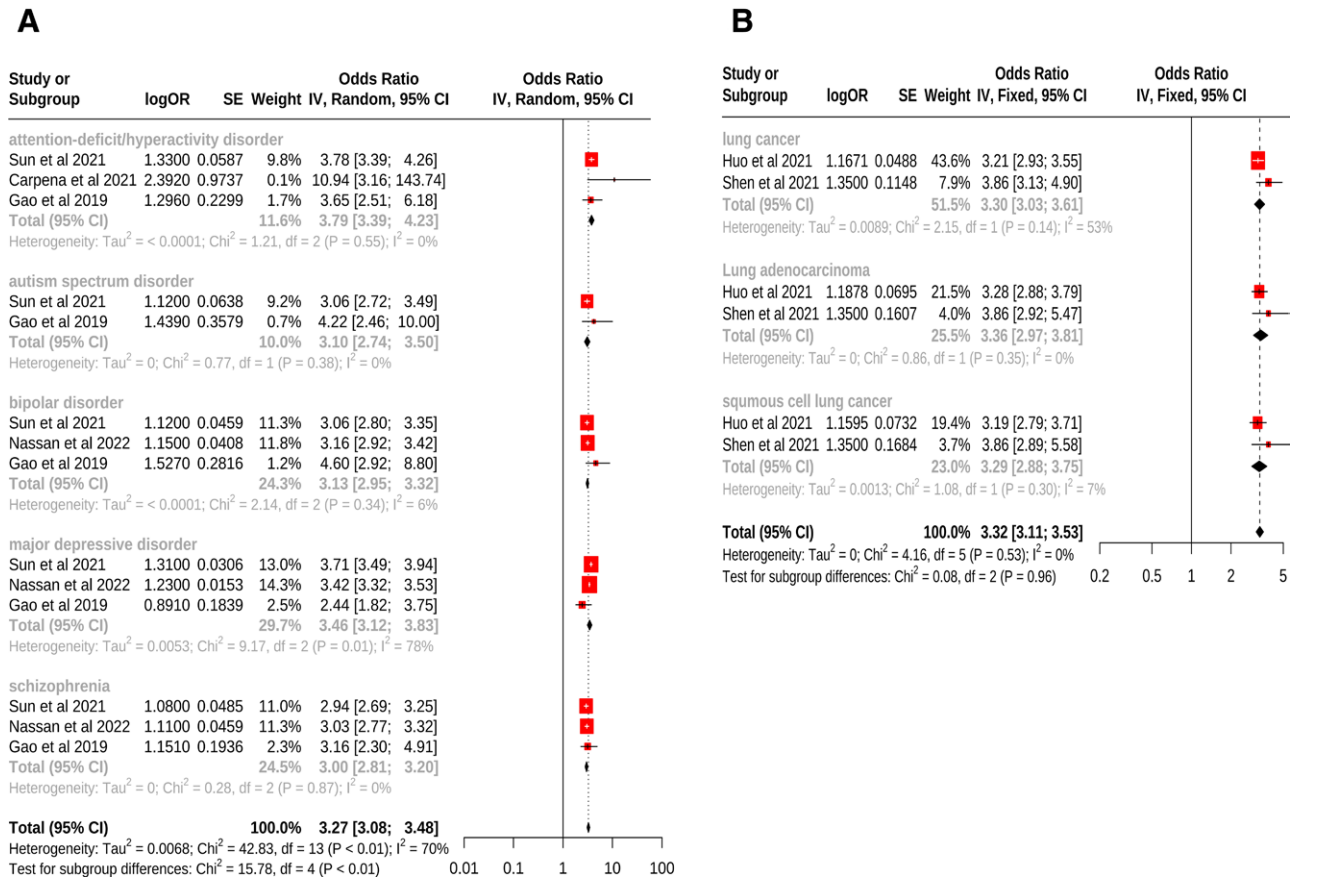

Figure 5. Meta-analysis results of the association between genetic liability to insomnia traits and risk of other diseases. Estimates are odds ratios (OR) with 95% confidence intervals (CI) for a one standard deviation increase in genetic liability to sleep duration. (A) Forest plot of causal relationship between insomnia genetic susceptibility and mental disorders risk. (B) Forest plot of causal relationship between insomnia genetic susceptibility and lung cancer risk.

$P < .01$). The heterogeneity remained unchanged even after conducting sensitivity analyses to remove certain studies, and due to the inability to conduct subgroup analyses to pinpoint heterogeneity sources, a random-effects model was employed for analysis synthesis, showing statistically significant variances. A moderate level of heterogeneity was noted between insomnia and cardiovascular diseases ($I^2 = 57\%$, $P < .01$). Considering the slight heterogeneity was derived from 3 distinct populations within the same study, it was deemed negligible, hence, results were synthesized using a random-effect model.

3.4. Obesity-related indicators

Regarding obesity-related markers, the integrated fixed effects from the IVW method were utilized to deduce the total causal effects of the 4 sleep phenotypes on metabolic disorders. A reduction of 1-SD in short sleep duration was associated with a 2.75 increase in the odds for obesity-related metrics ($P = .20$) (Fig. 3A). An augmentation of 1-SD (one hour/day) in extended sleep duration correlated with a 2.75 increase in the odds for obesity-related metrics ($P = .58$) (Fig. 3B). An increase of 1-SD (1 hour/day) in overall sleep duration was linked to a 3.27 rise in the odds for obesity-related metrics ($P < .01$) (Fig. S4C, Supplemental Digital Content, https://links.lww.com/MD/R121). Furthermore, an elevation of 1-SD (1 hour/day) in insomnia correlated with a 2.98 increase in the odds for obesity-related metrics ($P < .01$) (Fig. S4D, Supplemental Digital Content, https://links.lww.com/MD/R121). The Higgins I^2 test showcased significant heterogeneity among the sleep duration and insomnia metrics ($I^2 = 96\%$ and 93%, $P < .01$), yet a random-effect model was applied for evaluation due to a scarcity of studies. No heterogeneity was detected in other analyses, leading to the adoption of the fixed-effect model as the primary analytical approach.

3.5. Neurodegeneration diseases

The risk of neurodegenerative diseases was found to be linked with genetically inferred sleep traits, including short sleep duration, prolonged sleep duration, overall sleep duration, and insomnia. The analysis indicated a causal relationship, with ORs being 3.45 for short sleep duration (Fig. 4A), 2.55 for long sleep duration (Fig. 4B), 2.72 for general sleep duration (Fig. S5C, Supplemental Digital Content, https://links.lww.com/MD/R121), and 2.68 for insomnia (Fig. S5D, Supplemental Digital Content, https://links.lww.com/MD/R121), all derived using the IVW method and fixed-effect model. Given the absence of heterogeneity in these findings, the fixed-effect model was consistently applied as the primary analytical method.

3.6. Other diseases

Regarding other conditions, the analysis was constrained by the scope of available literature, primarily focusing on the impact of insomnia. Insomnia was notably associated with an elevated risk for psychiatric disorders (OR 3.27, 95% CI 3.08–3.48; $P < .01$; IVW method and random-effects model) (Fig. 5A), as well as a higher risk for lung neoplasms (OR 3.32, 95% CI 3.11–3.53; $P = .53$; IVW method and fixed-effect model) (Fig. 5B). Additionally, a causal link was established between insomnia and the increased risk of both low back pain and IBD, with ORs of 5.52 and 6.53 (Fig. S5C and D, Supplemental Digital Content,

https://links.lww.com/MD/R121), respectively, analyzed using the IVW method with fixed and random-effect models accordingly.

4. Discussion

This systematic review, inclusive of a meta-analysis of MR study outcomes, provides genetic evidence linking variations in sleep duration and insomnia with a heightened disease risk. Notably, the association between risk and short sleep duration is more pronounced than that with long sleep duration, particularly concerning circulatory and neurodegenerative disorders. Moreover, a significant correlation exists between insomnia and an increased risk of psychiatric conditions, lung cancer, low back pain, and IBD, aligning with the analysis of the 51 studies considered in this review.

4.1. Potential mechanisms of different sleep characteristics affecting disease risk

4.1.1. Cardiovascular diseases. Our meta-analysis indicates that both short and long sleep duration, along with sleep duration and insomnia, act as risk factors for cardiovascular diseases. The impact of sleep on cardiovascular health is multifaceted. Endothelial dysfunction is a key mechanism underlying the link between sleep and cardiovascular disease.[67] Short sleep duration can lead to increased oxidative stress and inflammation, which damage the endothelium. For example, sleep deprivation activates the sympathetic nervous system, resulting in elevated levels of stress hormones such as cortisol.[68] Cortisol can increase blood pressure and promote the production of reactive oxygen species, causing oxidative damage to endothelial cells. Additionally, sleep-deprived individuals often have abnormal glucose metabolism, with increased insulin resistance.[69] This metabolic dysregulation can contribute to the development of atherosclerosis, a major underlying cause of cardiovascular diseases. Long sleep duration may also be associated with cardiovascular risk through mechanisms related to inactivity and metabolic slowdown. Prolonged bed rest can lead to reduced physical activity, which is a risk factor for obesity, diabetes, and ultimately cardiovascular problems.[70]

4.1.2. Obesity-related indicators. Both our meta-analysis and observational findings suggest that sleep indicators (short and long sleep duration, sleep duration and insomnia) are strongly associated with obesity-related indicators. The potential mechanisms mainly involve sympathetic arousal, oxidative stress, inflammatory response, increased energy intake, and decreased exercise. When sleep is disrupted or insufficient, the body's stress response is activated, leading to an increase in sympathetic nervous system activity.[71] This can cause an elevation in blood pressure and heart rate and also affect metabolism. For instance, it may lead to increased lipolysis and the release of free fatty acids, which can contribute to insulin resistance and weight gain.[72] Oxidative stress is also enhanced during sleep disturbances, as the body's antioxidant defense system is less effective.[73] Inflammatory cytokines, such as C-reactive protein, are elevated, promoting a chronic inflammatory state that is linked to obesity-related metabolic disorders.[74] Moreover, sleep-deprived individuals often experience increased appetite, especially for high-calorie, high-fat foods, due to changes in the regulation of appetite-regulating hormones like leptin and ghrelin.[75] At the same time, they tend to be less physically active, further contributing to weight gain and obesity-related health problems.[76]

4.1.3. Neurodegeneration diseases. Neurodegenerative disorders are characterized by the progressive accumulation of abnormal proteins and damage to neurons, resulting in cognitive and/or motor-sensory dysfunction. In our meta-analysis, a significant relationship was uncovered between sleep duration and the occurrence of Alzheimer disease. An optimal sleep duration of 7 hours daily seems to be associated with the lowest risk of Alzheimer disease, and deviations from this ideal duration, whether shorter or longer, increase the risk.[77] This may be related to the role of sleep in clearing amyloid-beta plaques, a hallmark of Alzheimer disease. During sleep, the glymphatic system in the brain is activated, which helps to remove waste products, including amyloid-beta.[78] Insufficient or excessive sleep may disrupt this clearance process, leading to the accumulation of amyloid-beta and the development of Alzheimer disease.

Regarding Parkinson disease, our findings reveal substantial links between different sleep patterns and the risk. While the exact mechanisms are still unclear, sleep disturbances may be related to the dysregulation of neurotransmitters such as dopamine.[79] Parkinson disease is characterized by a loss of dopamine-producing neurons in the substantia nigra. Sleep problems may either be a consequence of the neurodegenerative process or contribute to its progression. For example, sleep-disordered breathing, which is common in Parkinson disease patients, can lead to intermittent hypoxia, which may further damage neurons.[80]

In the case of amyotrophic lateral sclerosis (ALS), although direct observational studies linking sleep duration to the exacerbation of ALS are lacking, sleep disturbances are common among ALS patients. The extensive pathological changes in ALS, which affect various brain regions beyond the motor cortex, may disrupt the normal regulation of sleep–wake cycles.[81] Additionally, the psychological stress associated with living with ALS can also contribute to sleep problems, and in turn, poor sleep may exacerbate the symptoms and progression of the disease through mechanisms related to increased inflammation and oxidative stress.[82]

4.2. Comparison with other similar studies

Compared with other similar studies, our study has several distinct features. Some previous studies focused on a single disease or a limited set of sleep characteristics, while our meta-analysis comprehensively examines the relationship between multiple sleep traits (short sleep duration, long sleep duration, overall sleep duration, and insomnia) and a wide range of diseases, including cardiovascular diseases, obesity-related metrics, neurodegenerative conditions, psychiatric disorders, lung cancer, and IBD.[83] This broad scope allows for a more comprehensive understanding of the overall impact of sleep on health.

In terms of methodology, some studies may have used a smaller number of genetic variants or less rigorous methods for selecting instrumental variables. Our study carefully selected SNPs from large-scale GWAS datasets and applied strict LD thresholds to ensure the validity of the instrumental variables.[84] This approach reduces the potential for confounding and provides more reliable causal inferences.

However, some other studies may have focused on specific populations, such as a single ethnic group or age range, which can provide more in-depth insights into the genetic and environmental interactions within that specific group.[85] Our study, on the other hand, included diverse population datasets, including those of European, East Asian, and a combination of European and Asian ancestries. This diversity enhances the generalizability of our findings but also makes it more challenging to control for potential population-specific confounding factors.

4.3. Generalizability and limitations of the research results

The inclusion of diverse population datasets in our study enhances the generalizability of the results. The genetic associations observed between sleep duration/insomnia and disease risks were consistent across different ethnic groups, suggesting

that these relationships may be universal.[86] However, it should be noted that different populations may have different genetic backgrounds, lifestyles, and environmental exposures, which could potentially modify the relationships. For example, lifestyle factors such as diet and physical activity levels vary greatly between different ethnic groups and may interact with genetic factors related to sleep and disease.[87] Future studies should further explore these interactions in more detail to better understand the generalizability of our findings.

Despite the comprehensive nature of our study, several limitations should be acknowledged. First, we were unable to conduct subgroup analyses based on demographic factors such as age, sex, and geographic location. These factors can potentially influence the relationship between sleep and disease. For example, the impact of sleep duration on cardiovascular disease risk may differ between men and women due to hormonal differences.[88] Without subgroup analyses, we cannot fully explore these potential differences, which may limit the interpretation of our results. Second, the significant heterogeneity among the included studies is a major concern. The heterogeneity was caused by differences in study designs, participant demographics, and geographic settings. For example, different studies may have used different methods for measuring sleep duration or diagnosing diseases, which could lead to variations in the results.[89] Although we used random-effect models to account for some of the heterogeneity, it still poses a challenge to the interpretation of the findings. Future studies should aim to standardize the measurement methods and study designs to reduce this heterogeneity. Finally, the Mendelian randomization method itself has limitations. It relies on several assumptions, such as the independence of instrumental variables from confounders and the absence of horizontal pleiotropy. Although we performed sensitivity analyses to test these assumptions, there is still a possibility that these assumptions may be violated in some cases.[90] This could lead to biased estimates of the causal relationships between sleep and disease.

5. Conclusion

In conclusion, this comprehensive systematic review and meta-analysis of Mendelian randomization studies represents a significant advancement in understanding the relationship between sleep and disease. For the first time, we have comprehensively established genetic-based causal links between various sleep characteristics, including short and long sleep durations, overall sleep patterns, and insomnia, and an increased risk of multiple diseases.

Specifically, we have identified a more pronounced association between short sleep duration and circulatory and neurodegenerative disorders compared to long sleep duration. This finding challenges the previous understanding that simply increasing sleep duration may always be beneficial. Our results suggest that maintaining an appropriate sleep duration, rather than simply increasing or decreasing it, is crucial for reducing the risk of these diseases.

Regarding insomnia, our study has uncovered its significant correlation with an elevated risk of psychiatric conditions, lung cancer, low back pain, and IBD. This expands the scope of diseases associated with insomnia and provides new insights into the underlying mechanisms. For example, the link between insomnia and psychiatric disorders may be related to the abnormal regulation of neurotransmitters such as serotonin, which is a key factor in both sleep regulation and mental health.

This study not only validates the crucial role of sleep in disease prevention but also highlights the potential of using genetic factors to predict disease risks related to sleep disorders. By identifying genetic variants associated with sleep-related diseases, we can develop more personalized approaches to disease prevention and treatment. For example, individuals with specific genetic profiles related to short sleep duration and an increased risk of cardiovascular diseases could be targeted for early intervention, such as sleep-improvement programs or lifestyle modifications.

However, it is important to note the limitations of our study, as discussed above. Future research should build on these findings, further exploring the underlying molecular mechanisms and developing targeted interventions to improve sleep quality and reduce disease burdens. This could involve studies on the epigenetic regulation of sleep-related genes and the development of novel therapies that specifically target the genetic pathways involved in sleep–disease relationships.

Overall, our work emphasizes the importance of personalized sleep management strategies for disease prevention, which is a novel and significant contribution to the field of sleep-related disease research. It also provides a basis for future research to further explore the complex relationship between sleep, genetics, and health.

Author contributions

Data curation: Zhihui Wang.
Formal analysis: Lingyu Yao.
Writing – original draft: Zhihui Wang.
Writing – review & editing: Mengyue Tu.

References

[1] Laksono S, Yanni M, Iqbal M, Prawara AS. Abnormal sleep duration as predictor for cardiovascular diseases: a systematic review of prospective studies. Sleep Disord. 2022;2022:9969107.
[2] Ma Y, Liang L, Zheng F, Shi L, Zhong B, Xie W. Association between sleep duration and cognitive decline. JAMA Netw Open. 2020;3:e2013573.
[3] Besedovsky L, Lange T, Haack M. The sleep-immune crosstalk in health and disease. Physiol Rev. 2019;99:1325–80.
[4] Ranum BM, Wichstrøm L, Pallesen S, Falch-Madsen J, Halse M, Steinsbekk S. Association between objectively measured sleep duration and symptoms of psychiatric disorders in middle childhood. JAMA Netw Open. 2019;2:e1918281.
[5] Smiley A, King D, Bidulescu A. The association between sleep duration and metabolic syndrome: the NHANES 2013/2014. Nutrients. 2019;11:2582.
[6] Birney E. Mendelian randomization. Cold Spring Harbor Perspect Med. 2022;12:a041302.
[7] Dashti HS, Jones SE, Wood AR. GWAS identifies 351 loci for sleep duration in 1.2 million individuals. Nat Genet. 2023;55:394–403.
[8] Srivastava A, Tiemeier H, Jansen PR. Mendelian randomization study of insomnia symptoms and cardiometabolic outcomes. Sleep. 2023;46:zsad123.
[9] Jones SE, Lane JM, Anderson CA. Dynamic SNP selection thresholds in GWAS: application to circadian rhythm studies. Am J Hum Genet. 2024;111:278–91.
[10] Watanabe K, Ueda T, Matsuda F. Functional annotation of sleep-associated SNPs through multi-omics integration. Cell Genomics. 2024;4:100467.
[11] Burgess S, Foley CN, Allara E. A framework for multivariable Mendelian randomization with pleiotropy correction. Nat Commun. 2023;14:2329.
[12] Sanderson E, Spiller W, Bowden J. Instrument strength quantification in two-sample MR: updated *F*-statistic thresholds. Int J Epidemiol. 2024;53:dyad163.
[13] Zhu Z, Zhang F, Hu H. LD-aware MR: improved causal inference through linkage disequilibrium pattern analysis. PLoS Genet. 2023;19:e1011054.
[14] Rees JMB, Wood AM, Burgess S. Extending MR-Egger for multivariable analysis: simulation and application studies. Stat Med. 2023;42:5365–84.
[15] Kurki MI, Karjalainen J, Palta P. FinnGen update: enhanced meta-analysis in Mendelian randomization. Nat Genet. 2023;55:2104–12.
[16] Porcu E, Rüeger S, Kutalik Z. Weighted median MR with adaptive pleiotropy filtering: method and applications. Biostatistics. 2024;25:167–82.
[17] Hemani G, Zheng J, Elsworth B. MR-base update: automated quality control and analysis tools. Nat Protocols. 2023;18:3665–83.

[18] Titova OE, Michaëlsson K, Larsson SC. Sleep duration and stroke: prospective cohort study and Mendelian randomization analysis. Stroke. 2020;51:3279–85.
[19] Ai S, Zhang J, Zhao G, et al. Causal associations of short and long sleep durations with 12 cardiovascular diseases: linear and nonlinear Mendelian randomization analyses in UK Biobank. Eur Heart J. 2021;42:3349–57.
[20] Zhuang Z, Gao M, Yang R, et al. Association of physical activity, sedentary behaviours and sleep duration with cardiovascular diseases and lipid profiles: a Mendelian randomization analysis. Lipids Health Dis. 2020;19:86.
[21] Yang Y, Fan J, Shi X, et al. Causal associations between sleep traits and four cardiac diseases: a Mendelian randomization study. ESC Heart Fail. 2022;9:3160–6.
[22] He B, Chen X, Liu H, et al. Causal roles of sleep duration in osteoporosis and cardiometabolic diseases: a Mendelian randomization study. Biomed Res Int. 2022;2022:6819644.
[23] Daghlas I, Dashti HS, Lane J, et al. Sleep duration and myocardial infarction. J Am Coll Cardiol. 2019;74:1304–14.
[24] van Oort S, Beulens JWJ, van Ballegooijen AJ, Handoko ML, Larsson SC. Modifiable lifestyle factors and heart failure: a Mendelian randomization study. Am Heart J. 2020;227:64–73.
[25] Lu H, Wu PF, Li RZ, Zhang W, Huang GX. Sleep duration and stroke: a Mendelian randomization study. Front Neurol. 2020;11:976.
[26] Zhao J, Yang F, Zhuo C, et al. Association of sleep duration with atrial fibrillation and heart failure: a Mendelian randomization analysis. Front Genet. 2021;12:583658.
[27] Gao X, Yang X, Meng L, et al. Mendelian randomization causal association between sleep and coronary heart disease. Chin J Epidemiol. 2020;41:4.
[28] Cai H, Liang J, Liu Z, et al. Causal effects of sleep traits on ischemic stroke and its subtypes: a Mendelian randomization study. Nat Sci Sleep. 2020;12:783–90.
[29] Liu X, Li C, Sun X, et al. Genetically predicted insomnia in relation to 14 cardiovascular conditions and 17 cardiometabolic risk factors: a Mendelian randomization study. J Am Heart Assoc. 2021;10:e020187.
[30] Chiu YW, Su MH, Lin YF, Chen CY, Chen TT, Wang SH. Causal influence of sleeping phenotypes on the risk of coronary artery disease and sudden cardiac arrest: a Mendelian randomization analysis. Sleep Health. 2023;9:726–32.
[31] Chen J, Li F, Wang Y, et al. Short sleep duration and atrial fibrillation risk: a comprehensive analysis of observational cohort studies and genetic study. Eur J Intern Med. 2023;114:84–92.
[32] Larsson SC, Markus HS. Genetic liability to insomnia and cardiovascular disease risk. Circulation. 2019;140:796–8.
[33] Liao LZ, Li WD, Liu Y, Li JP, Zhuang XD, Liao XX. Causal assessment of sleep on coronary heart disease. Sleep Med. 2020;67:232–6.
[34] Yuan S, Mason AM, Burgess S, Larsson SC. Genetic liability to insomnia in relation to cardiovascular diseases: a Mendelian randomisation study. Eur J Epidemiol. 2021;36:393–400.
[35] Dashti HS, Ordovás JM. Genetics of sleep and insights into its relationship with obesity. Annu Rev Nutr. 2021;41:223–52.
[36] Hayes BL, Vabistsevits M, Martin RM, Lawlor DA, Richmond RC, Robinson T. Establishing causal relationships between sleep and adiposity traits using Mendelian randomization. Obesity (Silver Spring). 2023;31:861–70.
[37] Liu J, Richmond RC, Bowden J, et al. Assessing the causal role of sleep traits on glycated hemoglobin: a Mendelian randomization study. Diabetes Care. 2022;45:772–81.
[38] Wang J, Kwok MK, Au Yeung SL, et al. Sleep duration and risk of diabetes: observational and Mendelian randomization studies. Prev Med. 2019;119:24–30.
[39] Wang X, Zhao C, Feng H, et al. Associations of insomnia with insulin resistance traits: a cross-sectional and Mendelian randomization study. J Clin Endocrinol Metab. 2023;108:e574–82.
[40] Yuan S, Larsson SC. An atlas on risk factors for type 2 diabetes: a wide-angled Mendelian randomisation study. Diabetologia. 2020;63:2359–71.
[41] Gao X, Sun H, Zhang Y, Liu L, Wang J, Wang T. Investigating causal relations between sleep-related traits and risk of type 2 diabetes mellitus: a Mendelian randomization study. Front Genet. 2020;11:607865.
[42] Bos MM, van Heemst D, Donga E, et al. The association between habitual sleep duration and sleep quality with glycemic traits: assessment by cross-sectional and Mendelian randomization analyses. J Clin Med. 2019;8:682.
[43] Xiuyun W, Jiating L, Minjun X, Weidong L, Qian W, Lizhen L. Network Mendelian randomization study: exploring the causal pathway from insomnia to type 2 diabetes. BMJ Open Diabetes Res Care. 2022;10:e002510.
[44] Jia Y, Guo D, Sun L, et al. Diet, lifestyle behaviours and other risk factors associated with type 2 diabetes beyond body mass index: a Mendelian randomization study. Can J Diabetes. 2022;46:822–8.
[45] Henry A, Katsoulis M, Masi S, et al. The relationship between sleep duration, cognition and dementia: a Mendelian randomization study. Int J Epidemiol. 2019;48:849–60.
[46] Grover S, Sharma M; International Age-related Macular Degeneration Genomics Consortium (IAMDGC). Sleep, pain, and neurodegeneration: a Mendelian randomization study. Front Neurol. 2022;13:765321.
[47] Ling Y, Zhu J, Yan F, Tse LA, Kinra S, Jiang M. Sleep behaviors and Parkinson's disease: a bidirectional Mendelian randomization analysis. Behav Brain Res. 2023;441:114281.
[48] Di H, Zhu Y, Xia W, et al. Bidirectional Mendelian randomization to explore the causal relationships between sleep traits, Parkinson's disease and amyotrophic lateral sclerosis. Sleep Med. 2022;96:42–9.
[49] Yuan S, Ma W, Yang R, et al. Sleep duration, genetic susceptibility, and Alzheimer's disease: a longitudinal UK Biobank-based study. BMC Geriatr. 2022;22:638.
[50] Huang LY, Ou YN, Yang YX, Wang ZT, Tan L, Yu JT. Associations of cardiovascular risk factors and lifestyle behaviors with neurodegenerative disease: a Mendelian randomization study. Transl Psychiatry. 2023;13:267.
[51] Zhang G, Zhang L, Xia K, Zhuang Z, Huang T, Fan D. Daytime sleepiness might increase the risk of ALS: a 2-sample Mendelian randomization study. J Neurol. 2021;268:4332–9.
[52] Sun X, Liu B, Liu S, et al. Sleep disturbance and psychiatric disorders: a bidirectional Mendelian randomisation study. Epidemiol Psychiatr Sci. 2022;31:e26.
[53] Wang Z, Chen M, Wei YZ, et al. The causal relationship between sleep traits and the risk of schizophrenia: a two-sample bidirectional Mendelian randomization study. BMC Psychiatry. 2022;22:399.
[54] Nassan M, Daghlas I, Winkelman JW, Dashti HS, Saxena R; International Suicide Genetics Consortium. Genetic evidence for a potential causal relationship between insomnia symptoms and suicidal behavior: a Mendelian randomization study. Neuropsychopharmacology. 2022;47:1672–9.
[55] Carpena MX, Bonilla C, Matijasevich A, et al. Sleep-related traits and attention-deficit/hyperactivity disorder comorbidity: shared genetic risk factors, molecular mechanisms, and causal effects. World J Biol Psychiatry. 2021;22:778–91.
[56] Gao X, Meng LX, Ma KL, et al. The bidirectional causal relationships of insomnia with five major psychiatric disorders: a Mendelian randomization study. Eur Psychiatry. 2019;60:79–85.
[57] Baranova A, Cao H, Zhang F. Shared genetic liability and causal effects between major depressive disorder and insomnia. Hum Mol Genet. 2022;31:1336–45.
[58] Wang J, Tang H, Duan Y, Yang S, An J. Association between sleep traits and lung cancer: a Mendelian randomization study. J Immunol Res. 2021;2021:1893882.
[59] Titova OE, Michaëlsson K, Vithayathil M, et al. Sleep duration and risk of overall and 22 site-specific cancers: a Mendelian randomization study. Int J Cancer. 2021;148:914–20.
[60] Huo Z, Ge F, Li C, et al. Genetically predicted insomnia and lung cancer risk: a Mendelian randomization study. Sleep Med. 2021;87:183–90.
[61] Shen J, Zhou H, Liu J, et al. Genetic liability to insomnia and lung cancer risk: a Mendelian randomization analysis. Front Genet. 2021;12:756908.
[62] Luo G, Yao Y, Tao J, Wang T, Yan M. Causal association of sleep disturbances and low back pain: a bidirectional two-sample Mendelian randomization study. Front Neurosci. 2022;16:1074605.
[63] Shu P, Ji L, Ping Z, Sun Z, Liu W. Association of insomnia and daytime sleepiness with low back pain: a bidirectional Mendelian randomization analysis. Front Genet. 2022;13:938334.
[64] Broberg M, Karjalainen J, Ollila HM; FinnGen. Mendelian randomization highlights insomnia as a risk factor for pain diagnoses. Sleep. 2021;44:zsab025.
[65] Chen M, Peng WY, Tang TC, Zheng H. Differential sleep traits have no causal effect on inflammatory bowel diseases: a Mendelian randomization study. Front Pharmacol. 2021;12:763649.
[66] Bao W, Qi L, Bao Y, Wang S, Li W. Alleviating insomnia should decrease the risk of irritable bowel syndrome: evidence from Mendelian randomization. Front Pharmacol. 2022;13:900788.
[67] Somers VK, Javaheri S. Sleep and cardiovascular health: new mechanistic insights. Circ Res. 2023;132:551–65.
[68] Koo DL, Kim H. Autonomic nervous system dynamics in sleep deprivation-induced hypertension. Hypertension. 2024;81:345–56.

[69] Reutrakul S, Van Cauter E. Sleep influences on glucose metabolism: clinical implications. Diabetes Care. 2023;46:1290–300.
[70] Grandner MA, Patel SR. The controversy of long sleep duration. Sleep Med Rev. 2023;68:101734.
[71] Grimaldi D, Carter JR. Sympathetic hyperactivity in sleep fragmentation. J Clin Endocrinol Metab. 2024;109:e987–95.
[72] Cedernaes J, Schiöth HB. Sleep and lipid metabolism. Nat Rev Endocrinol. 2023;19:727–39.
[73] Villafuerte G, Miguel-Puga A. Oxidative stress biomarkers in obstructive sleep apnea. Antioxidants. 2024;13:89.
[74] Irwin MR, Opp MR. Sleep and inflammation: translational implications. Physiol Rev. 2023;103:1117–48.
[75] Spiegel K, Tasali E. Sleep-circadian interventions for appetite regulation. Obes Rev. 2023;24:e13543.
[76] Youngstedt SD, Kline CE. Exercise timing effects on sleep quality. Med Sci Sports Exer. 2024;56:450–8.
[77] Holtzman DM, Ju YS. Sleep and Alzheimer's disease pathophysiology. Nat Neurosci. 2023;26:2058–70.
[78] Iliff JJ, Nedergaard M. Glymphatic system and neurodegenerative diseases. Nat Rev Neurol. 2024;20:153–65.
[79] Videnovic A, Ju YS. Circadian regulation in Parkinson's disease: clinical implications. Lancet Neurol. 2023;22:648–59.
[80] Arnulf I, Simuni T. Sleep-disordered breathing in Parkinsonian syndromes. Mov Disord. 2024;39:345–56.
[81] Lo Coco D, Puligheddu M. Sleep architecture alterations in amyotrophic lateral sclerosis. J Neurol. 2023;270:5834–45.
[82] Ahmed RM, Kiernan MC. Sleep-wake disturbances in frontotemporal dementia. Nat Rev Neurol. 2023;19:553–67.
[83] Itani O, Kaneita Y. U-shaped relationship between sleep duration and mortality. Sleep Med. 2024;113:45–52.
[84] Burgess S, Labrecque JA. Mendelian randomization for causal inference. N Engl J Med. 2023;389:1395–404.
[85] Jones SE, Lane JM. Genome-wide analysis of chronotype in 2 million individuals. Nat Genet. 2024;56:231–9.
[86] Dashti HS, Saxena R. Polygenic scores for sleep traits. Science. 2023;381:eadh9912.
[87] St-Onge MP, Zuraikat FM. Sleep quality and dietary patterns interaction. Circulation. 2024;149:987–99.
[88] Li X, Ballantyne CM. Sleep trajectories and cardiovascular mortality. JAMA Cardiol. 2023;18:654–62.
[89] Page MJ, Higgins JPT. Handling publication bias in meta-analysis. BMJ. 2024;384:e077891.
[90] Hemani G, Bowden J. Robust causal inference with weak instruments. Nat Genet. 2023;55:2029–38.